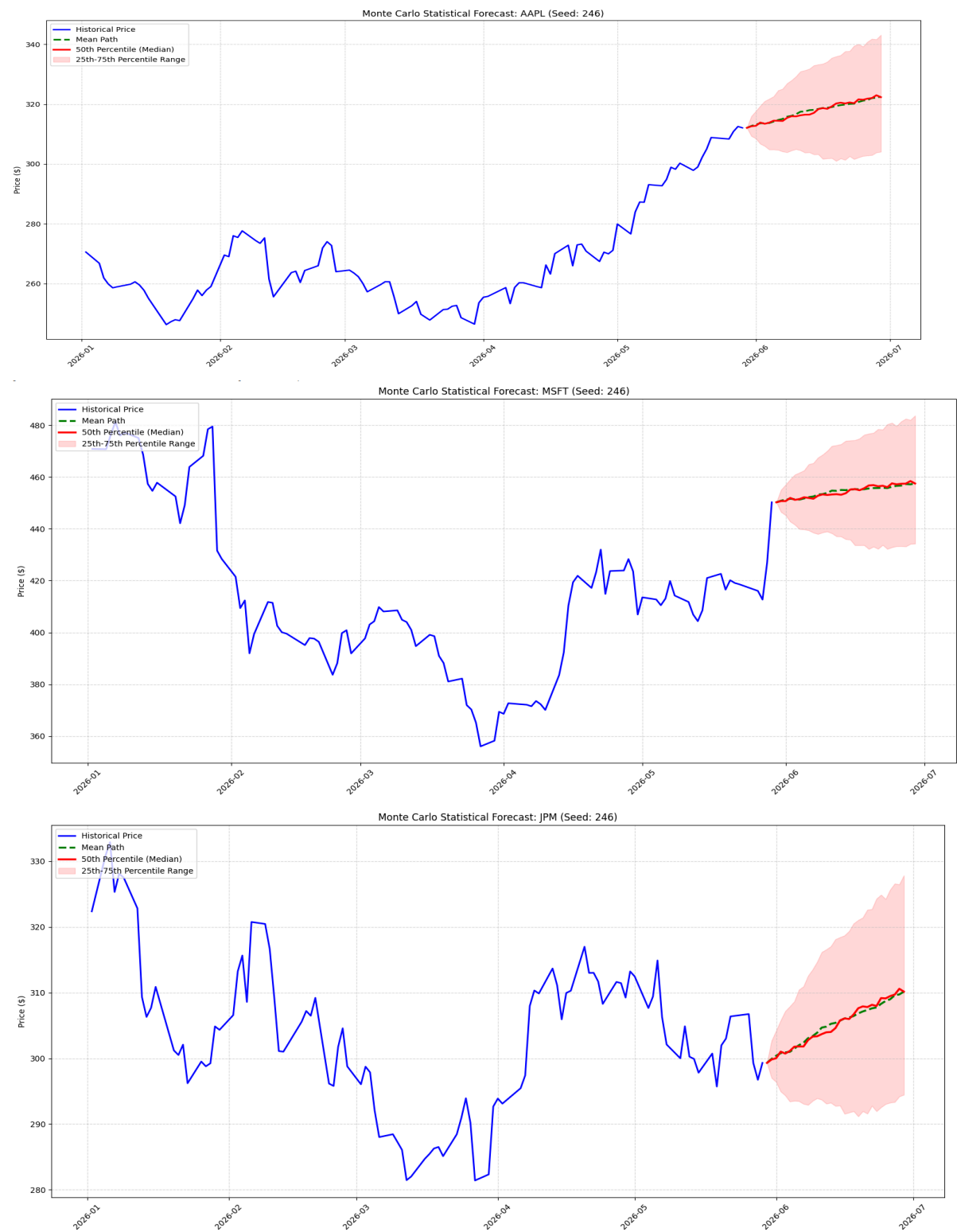
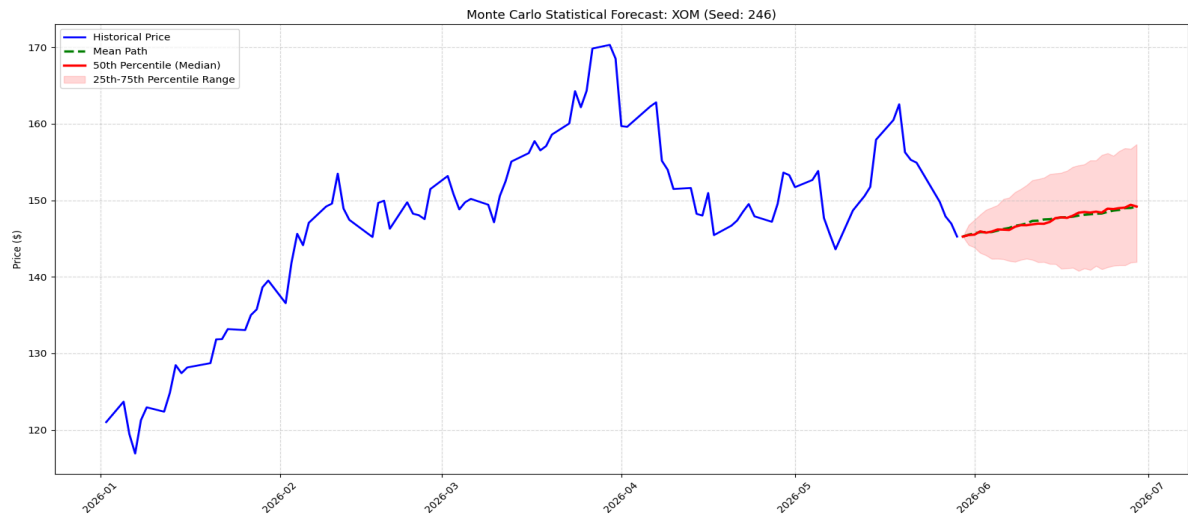


Portfolio Optimization II

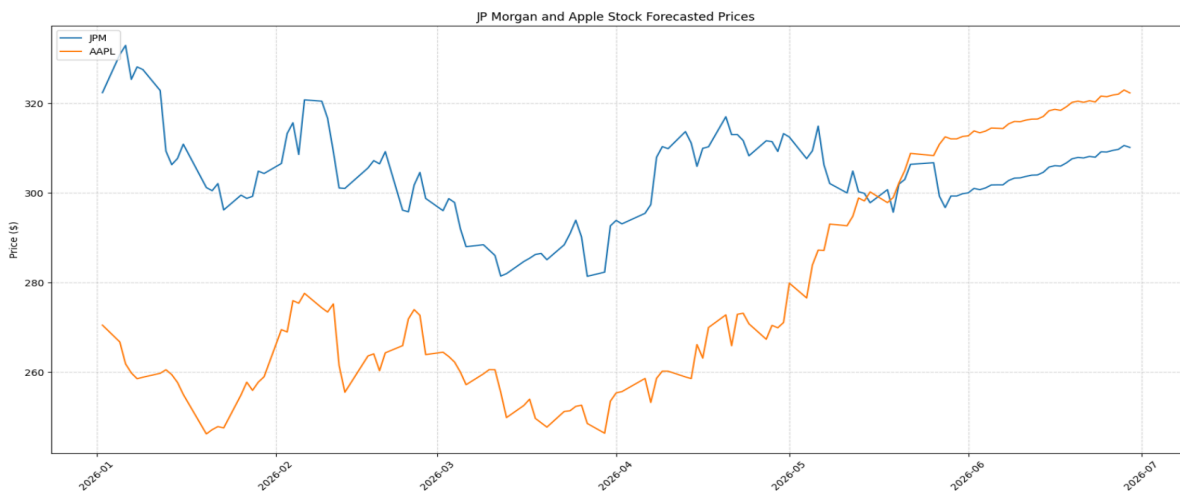
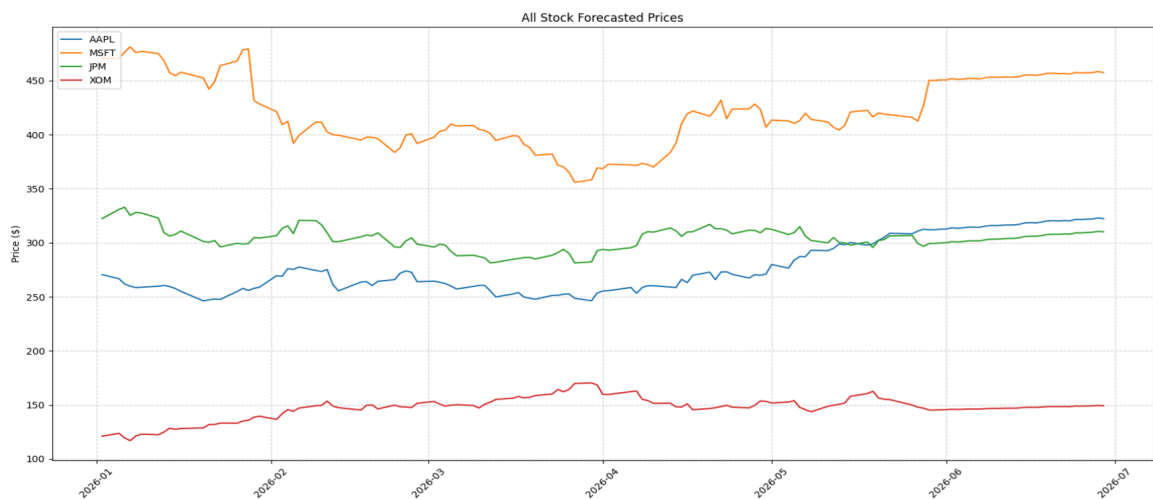
This document shows the forecasted paths for each stock for June using Monte Carlo and how each stock has performed.

Forecasted Paths



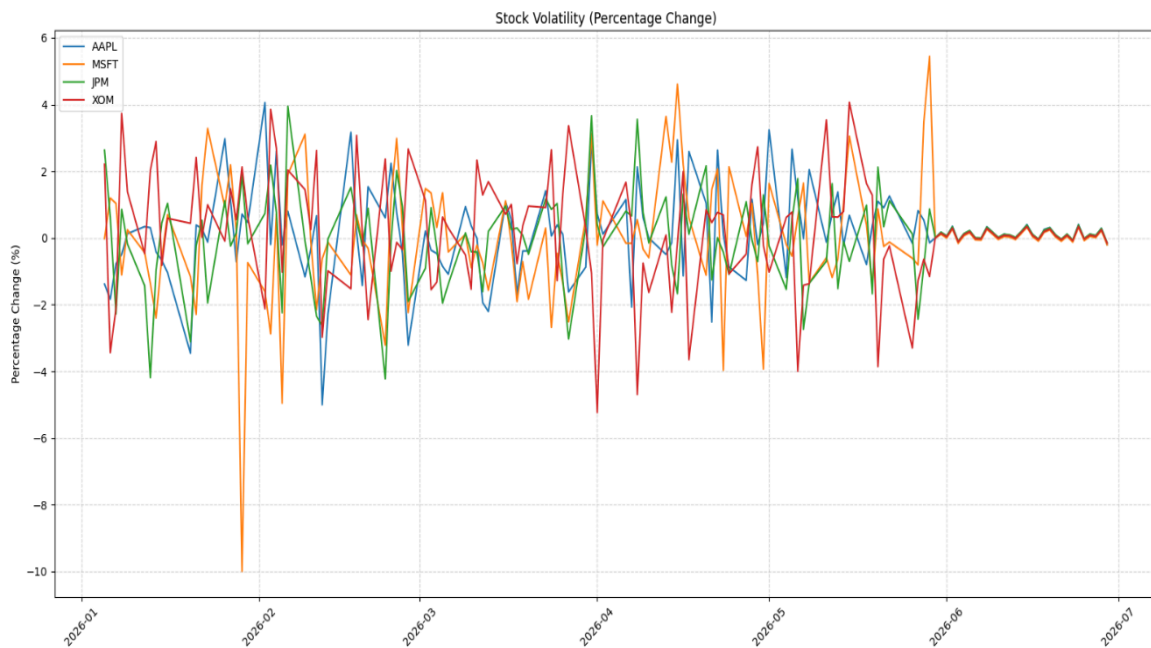


- Notable is the relatively low volatility of the 50th percentile paths with steady and linear trend similar to the mean path. The JP Morgan stock shows the most volatility in terms of the range of the 25th-75th percentile as its the widest. At the same time, by eye inspection it has the most growth within the month of June.



- The Apple stock price remains above JP Morgan with little to no volatility, this is highly unrealistic and may require further modelling in future.

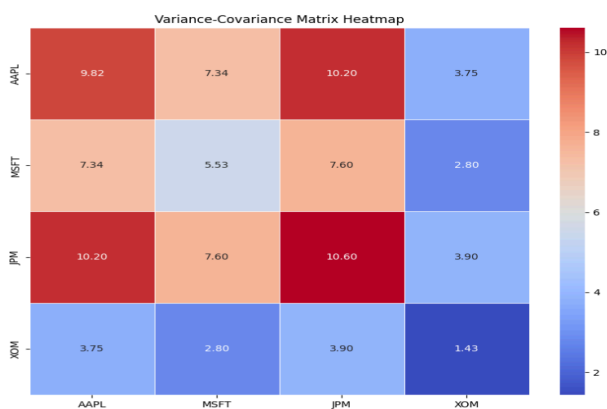
Forecasted Volatility



- Without isolating the volatility plots, one can see the low volatility in the forecasts is relatively low for 30 trading days for all the stocks. Scrutinising the volatility for the forecasts however shows mostly positive returns rather than negative.

	AAPL	MSFT	JPM	XOM
count	29.000000	29.000000	29.000000	29.000000
mean	317.899032	454.583413	305.384426	147.493756
std	3.133783	2.351062	3.256362	1.197417
min	312.733322	450.708802	300.013269	145.513965
25%	315.410223	452.784737	302.775773	146.543642
50%	318.335789	454.898091	305.768018	147.675433
75%	320.475817	456.747749	307.968150	148.477264
max	322.960013	458.438924	310.585574	149.415594

- The heatmap below shows that variance within and between forecasted values is very low with the highest being \$10.20.



Forecasted Prices Distributions

Each forecasted prices histogram also shows the closing price on 31 May 2026, the day on which the portfolio is initialized.

